

Getting started with Amazon Relational Database Service (Amazon RDS)

AWS Prescriptive Guidance

Amazon RDS makes it easy to set up, operate, and scale a relational database in the cloud. It provides cost-efficient, resizable capacity while automating time-consuming administration tasks such as hardware provisioning, database setup, patching, and backups. This guide covers RDS engine options, Multi-AZ, Read Replicas, Aurora, backup and recovery, performance insights, and database security best practices.

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Date: February 2026

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Introduction

Amazon RDS is a managed relational database service that automates the undifferentiated heavy lifting associated with database administration, including hardware provisioning, OS patching, database software patching, automated backups, and monitoring. Amazon RDS runs on the same highly reliable infrastructure used by other Amazon Web Services and is available across multiple AWS Regions and Availability Zones.

With Amazon RDS, you can scale database compute and storage resources independently with just a few mouse clicks or a simple API call. Many Amazon RDS database instance types are optimized to use SSDs and are optimized for memory, performance, or I/O. The service can manage multi-petabyte databases with confidence.

Supported Database Engines

Engine	Latest Version	Compatible With	Key Strengths
MySQL	8.0	MySQL 5.7, 8.0	Wide adoption, large ecosystem, web applications
PostgreSQL	16	PostgreSQL 12–16	Advanced SQL features, extensibility, JSONB support
MariaDB	10.11	MySQL 5.6, 5.7	Open-source MySQL fork, performance improvements
Oracle	19c, 21c	Oracle 12c, 19c, 21c	Enterprise features, licensing options, PL/SQL
SQL Server	2022	SQL Server 2012–2022	Windows workloads, .NET integration, SSRS/SSIS
Db2	11.5	Db2 11.5	IBM compatibility, mainframe migration

Amazon Aurora

Amazon Aurora is a MySQL- and PostgreSQL-compatible relational database built for the cloud that combines the performance and availability of traditional enterprise databases with the simplicity and cost-effectiveness of open source databases. Aurora delivers up to five times the throughput of standard MySQL and up to three times the throughput of standard PostgreSQL.

Aurora uses a distributed, fault-tolerant, self-healing storage system that auto-scales up to 128 TB per database instance. Aurora replicates six copies of your data across three Availability Zones and continuously backs up your data to Amazon S3. Aurora maintains separate storage volumes for compute and storage, allowing them to scale independently.

Aurora Serverless v2

Aurora Serverless v2 automatically scales capacity in fine-grained increments (0.5 ACU steps) based on application demand, within seconds. It is ideal for workloads with variable or intermittent traffic patterns, including development and test databases, new applications with uncertain demand, and databases with infrequent peaks. Aurora Serverless v2 supports Multi-AZ, read replicas, and Global Databases.

Aurora Global Database

Aurora Global Database is designed for globally distributed applications, allowing a single Aurora database to span multiple AWS Regions. It has minimal impact on the primary database, with replication performed asynchronously at the storage layer with typical latency under one second. Aurora Global Database supports up to five secondary Regions, each with up to 16 read replicas. In the event of a regional failure, you can promote a secondary Region to become the primary in under one minute.

High Availability with Multi-AZ

Amazon RDS Multi-AZ deployments provide enhanced availability and durability. When you provision a Multi-AZ DB instance, Amazon RDS automatically creates a primary DB instance and synchronously replicates the data to a standby instance in a different Availability Zone. In the event of a primary instance failure, infrastructure fault, or Availability Zone disruption, Amazon RDS performs an automatic failover to the standby, typically completing in 60 to 120 seconds.

Multi-AZ DB Clusters (available for MySQL 8.0 and PostgreSQL 13+) provide two readable standby DB instances in separate Availability Zones, with failover completing in under 35 seconds. Readable standbys can serve read traffic, improving read scalability without the replication lag of traditional read replicas.

Note: Multi-AZ deployments are for high availability, not for read scaling. Use Read Replicas to scale read workloads horizontally.

Read Replicas

Amazon RDS Read Replicas provide enhanced performance and durability for database instances. They make it easy to elastically scale out beyond the capacity constraints of a single DB instance for read-heavy database workloads. You can create up to 15 Aurora read replicas, 5 MySQL or MariaDB read replicas, and 5 PostgreSQL read replicas per primary instance.

Cross-Region Read Replicas allow you to direct read traffic to a replica in a different AWS Region, reducing read latency for globally distributed users. Cross-Region replicas can also be promoted to standalone instances as part of a disaster recovery strategy. Replication between regions uses secure, encrypted channels.

Backup and Recovery

Amazon RDS provides automated backups, database snapshots, and the ability to restore a DB instance to a specific point in time. Automated backups are enabled by default and allow you to recover your database to any point within your backup retention period, which can be set between 1 and 35 days. Backups are stored in Amazon S3, encrypted using AES-256.

Point-in-time recovery (PITR) allows you to restore your database to any second during your retention period, up to the latest restorable time (usually within 5 minutes of the current time). DB snapshots are user-initiated backups that persist until you explicitly delete them and can be copied across Regions and accounts.

Performance Insights

RDS Performance Insights is a database performance tuning and monitoring feature that visually assesses the load on your database. It helps you detect performance problems through intuitive dashboards showing database load by wait type, SQL statements, users, and hosts. The DB Load metric represents the average active sessions (AAS) in your database, and the dashboard allows you to identify which queries are consuming the most resources.

Performance Insights retains performance data for seven days free of charge. You can extend retention to up to two years by upgrading to the Performance Insights long-term retention tier. Performance Insights supports all RDS engines except Oracle when using the Bring Your Own License (BYOL) model.

Security

Amazon RDS provides multiple security mechanisms. Database instances run within a VPC, providing network isolation. Security groups control inbound and outbound access to DB instances. IAM database authentication allows you to authenticate to your DB instance using IAM users and roles instead of database passwords. All data at rest is encrypted using AES-256 via AWS KMS, and encryption must be configured at instance creation time.

Best Practices

- Always enable Multi-AZ for production workloads: Multi-AZ provides automatic failover and protects against data loss.
- Use RDS Proxy for connection pooling: RDS Proxy pools and shares database connections to improve application scalability and resilience.
- Enable automated backups: Set a backup retention period appropriate for your RPO requirements (1–35 days).

- Right-size your instance class: Use Performance Insights and CloudWatch metrics to identify over- or under-provisioned instances.
- Use IAM database authentication: Replace static database passwords with IAM-based authentication for improved security posture.
- Encrypt your database at rest and in transit: Enable encryption at instance creation and enforce SSL/TLS connections.
- Place RDS in private subnets: Never place RDS instances in public subnets; use application-layer security groups to control access.
- Test failover regularly: Perform regular failover tests to validate your Multi-AZ configuration and measure actual failover times.

Frequently Asked Questions

Q: When should I use Aurora instead of standard RDS?

A: Choose Aurora when you need higher performance, larger storage (up to 128 TB), more replicas, global distribution, or Aurora Serverless capabilities. Standard RDS may be preferred for specific engine version requirements or BYOL licensing scenarios.

Q: What is the difference between a Multi-AZ deployment and a read replica?

A: Multi-AZ provides high availability through synchronous replication to a standby instance in a different AZ — it is not for read scaling. Read replicas use asynchronous replication to provide read scalability and can be promoted to standalone instances.